

A Day in the Life of a Data Engineer

SPL-BE-200-ANDEB1-1 - Version 1.0.0

Lab overview

Your colleague is out on vacation, so you're in charge of your organization's data engineering practice for the day. Step into their shoes and explore various managed options for data movement on Amazon Web Services (AWS). Consider architecture patterns, performance, cost optimizations, and security best practices, and impress your colleague when they get back to the office!

Your colleague has provided you with a sandbox environment so that you can experiment and test out a proof of concept (POC) while they are away. You have decided to explore data that you believe should help your small company successfully expand into new cities. You use a National Oceanic and Atmospheric Administration (NOAA) dataset that provides you with historical daily weather data. You believe that this data can help your company determine when it should stock summer, as opposed to winter, items in cities that you are expanding into. For your POC, you focus on temperature and precipitation metrics.

This lab uses the **NOAA Global Surface Summary of Day (GSOD)** dataset, which was accessed on August 5, 2022, from [Registry of Open Data on AWS](#). This dataset is a collection of daily weather measurements (temperature, wind speed, humidity, pressure, and more) from 9000+ weather stations around the world. Data was originally collected by the National Climatic Data Center. Because of the lab time limit, the dataset used in this lab has been reduced to include a subset of the years that are available in the full dataset.

Objectives

By the end of this lab, you should be able to do the following:

- Create an AWS Glue crawler.
- Create and run a job in AWS Glue Studio.

- Explore permissions required to run AWS Glue crawlers and AWS Glue Studio jobs.
- Query the AWS Glue Data Catalog using Amazon Athena.

Technical knowledge prerequisites

This lab requires the following prerequisites:

- Access to a computer with Microsoft Windows, Mac OS X, or Linux (Ubuntu, SuSE, or Red Hat)
- A modern internet browser such as Chrome or Firefox
- Basic familiarity with cloud and AWS services

Icon key

- **Note:** A hint, tip, or important guidance.
- **Learn more:** Where to find more information.
- **Caution:** Information of special interest or importance (not important enough to cause problems with equipment or data if you miss it, but it could result in the need to repeat certain steps).
- **Refresh:** A time when you might need to refresh a web browser page or list to show new information.
- **Task complete:** A conclusion or summary point in the lab.

Start lab

1. To launch the lab, at the top of the page, choose **Start Lab**.
Caution: You must wait for the provisioned AWS services to be ready before you can continue.

2. To open the lab, choose **Open Console**.

You are automatically signed in to the AWS Management Console in a new web browser tab.

Warning: Do not change the **Region** unless instructed.

Common sign-in errors

Error: Choosing Start Lab has no effect

In some cases, certain pop-up or script blocker web browser extensions might prevent the **Start Lab** button from working as intended. If you experience an issue starting the lab:

- Add the lab domain name to your pop-up or script blocker's allow list or turn it off.
- Refresh the page and try again.

AWS services not used in this lab

AWS service capabilities used in this lab are limited to what the lab requires. Expect errors when accessing other services or performing actions beyond those provided in this lab guide.

Task 1: Create and run an AWS Glue crawler

In this task, you create an AWS Glue crawler to scan the data in an Amazon Simple Storage Service (Amazon S3) bucket, infer the schema of the data, and filter data that should be included in the Data Catalog.

Because you can access the GSOD dataset directly from Amazon S3, you can quickly start exploring it to find data. You do not need to create a relational or NoSQL database to query and prepare the data. Instead, you use AWS Glue.

AWS Glue is a serverless data integration service that you can use to discover, prepare, and combine data for analytics, machine learning (ML), and application development. AWS Glue provides all the capabilities needed for data integration so that you can start analyzing your data and putting it to use in minutes instead of months.

Learn more: Refer to *AWS Glue Documentation* in the **Additional resources** section for more information.

3. At the top of the AWS Management Console, in the search bar, search for and choose **AWS Glue**.

4. In the left navigation pane, under **Data Catalog**, choose **Crawlers**.
5. Choose **Create crawler**.
6. On the **Set crawler properties** step, configure the **Crawler details** as follows:
 - For **Name**, enter `weather_crawler`.
 - For **Description**, enter `Crawler to read data from Amazon S3`.
7. Choose **Next**.
8. On the **Choose data sources and classifiers** step, under **Data source configuration** section, for **Is your data already mapped to Glue tables?** choose **Not yet**.
9. Under **Data sources**, choose **Add a data source** and configure the following settings:
 - For **Data source**, use the dropdown menu and select **S3**.
 - For **Network connection**, keep this blank.

Note: You can safely ignore the below error for this lab:

Error fetching connections

- For **Location of S3 data**, choose **In a different account**.
- For **S3 path**, enter `s3://aws-tc-largeobjects/SPL-BE-200-ANDEB1-1/data/`.
- For **Subsequent crawler runs**, choose **Crawl all sub-folders**.
- Select **Exclude files matching pattern**.

Caution: You might need to choose the check box more than once before the button named **Add new exclusion pattern** appears.

Exclude patterns prevent the crawler from reading all data under the target location. Using exclude patterns helps speed up processing when you do not need to ingest all data in the target location.

Learn more: Refer to *Include and exclude patterns* in the **Additional resources** section for more information.

You add four exclusion patterns. Repeat the following process to add each pattern:

10. Choose **Add new exclusion pattern**.
 - For **Exclude pattern**, enter **19****.
11. Choose **Add new exclusion pattern** again and add below exclusion patterns:
 - **200****
 - **201[0-2]****
 - **index.html**
12. After you add all exclusion patterns, choose **Add an S3 data source**.
13. Choose **Next**.
14. On the **Configure security settings** step, under **IAM role** section, for **Existing IAM role**, use the dropdown menu and choose **gluelab**.
15. Choose **Next**.
16. On the **Set output and scheduling** step, under **Output configuration** section, for **Target database**, choose **Add database**.

This re-directs you to the **Create a database** page.

17. In the **Database details** pane, for **Name**, enter **weather_data**.
18. Choose **Create database**.
19. Close the **Databases** browser tab to return to the Glue crawler configuration.
20. Under **Output configuration**, next to the **Target database** dropdown list, choose the refresh icon.
21. For **Target database**, use the dropdown menu and choose **weather_data**.
22. For **Crawler schedule**, use the dropdown menu and choose **On demand**.
23. Choose **Next**.

24. On the **Review and create** step, review the configuration and then choose **Create crawler**.
25. In the left navigation pane, below **Data Catalog**, choose **Crawlers**.
26. If you don't see your new crawler, refresh the page.
27. In the **Crawlers** pane, choose **weather_crawler**.
28. Choose **Run crawler**.

The crawler **State** changes from *READY* to *RUNNING*. The crawler is now reading data from the S3 bucket to build a Data Catalog table. The crawler runs for about 5 minutes.

While the crawler runs, continue to the next task, but be sure to keep the **Crawlers** tab open.

Task complete: You have successfully created and run an AWS Glue crawler.

Task 2: Review the IAM policies

In this task, you explore the AWS Identity and Access Management (IAM) permissions that were used to run the AWS Glue processes.

While you wait for the data transformation job to complete, use the IAM console to review the *gluelab* role. This is the role that you chose when you ran the AWS Glue crawler. You also use this role later, when you run a data transformation job in AWS Glue Studio.

29. At the top of the AWS Management Console, in the search bar, search for **IAM**, right-click (or an equivalent UI interaction) on IAM, and choose **Open Link in New Tab**.
30. Choose the **IAM | Global** tab.
31. In the left navigation pane, under **Access management**, choose **Roles**.
32. In the search bar, enter **gluelab**.

33. Under **Role name**, choose **gluelab**.

Under the **Permissions** tab, you find two AWS managed policies that are associated with this role.

34. Choose **gluePolicyRestrictS3**.

A new browser tab opens.

35. On the Permissions tab, in the **Summary** pane, notice the policy definition.

This lab policy provides Amazon S3 with read and write access to the lab environment's S3 buckets. The AWS Glue job that you created requires these privileges because it reads and writes information to Amazon S3 while processing data.

36. Close the **gluePolicyRestrictS3** browser tab.

37. Choose **AWSGlueServiceRole**.

A new browser tab opens.

38. On the Permissions tab, in the **Summary** pane, notice the policy definition.

This lab policy gives AWS Glue access to related services such as Amazon Elastic Compute Cloud (Amazon EC2), Amazon S3, and Amazon CloudWatch Logs. With these privileges, AWS Glue can read, write, and process data. It can also write information to Amazon CloudWatch logs.

39. Close the **AWSGlueServiceRole** browser tab.

40. On the **gluelab** pane, choose the **Trust relationships** tab.

In this trust policy, the AWS Glue service is given access to assume the role. When AWS Glue assumes the role, it receives all privileges that are included in the role's policies.

41. Close the **gluelab | IAM | Global** tab and return to the browser tab where the crawler is running.

You now know the basic permissions that are required to run AWS Glue processes. Nicely done!

42. Verify that the crawler has completed successfully before you continue to the next task.

Wait for the crawler run to complete before proceeding to Task 3. When the job completes, the Crawler **State** changes to *READY*.

Under the **Crawler runs** tab, the **Status** column shows **Completed**.

Task complete: You have successfully reviewed the IAM permissions that were used to run the AWS Glue processes.

Task 3: View the table in the Data Catalog

In this task, you view the table that the AWS Glue crawler added to the Data Catalog. You explore the metadata that was generated by the crawler, and you locate the partitions that are associated with the table.

43. In the navigation pane, under **Data Catalog**, choose **Tables**.

You find an entry for a table named **data**. Notice that the **Classification** for this data is **CSV**. This classification was inferred by AWS Glue when it read the data from the source S3 bucket.

44. Choose **data**.

45. From the **Actions** menu, choose **View properties**.

From this page, you can view table metadata such as when it was last updated, the input format and delimiter, exclusions, and the table schema.

46. Close the **Properties** window.

47. Choose the **Partitions** tab to view a list of the table's partitions.

The lab dataset on Amazon S3 includes data from 1972 through 2022. However, you applied a filter to the AWS Glue crawler so your Glue table only contains data from 2013 through 2022.

Task complete: You have successfully learned how to locate the metadata for a Data Catalog table and how to find the table's partition details.

Task 4: Run a job in AWS Glue Studio to transform the data

In this task, you create and run an AWS Glue Studio job to clean up the data.

Using an AWS Glue crawler, you created a table in the Data Catalog. However, the data isn't quite ready to use yet. You discovered that one of the data types needs to change. You also found that many columns in this dataset are empty.

Task 4.1: Choose and configure the data source for the job

48. In the navigation pane at the left of the page, under **Data Integration and ETL**, choose **ETL jobs**.

The browser displays the **AWS Glue Studio** page.

49. In the **Create job** pane, choose **Visual ETL**.
50. On the **Visual** tab, in the **+ Add nodes** pane, choose the **Sources** tab.
51. From the list of available sources, choose **Amazon S3**.

An **Amazon S3** card is placed on the canvas.

52. Choose the **Amazon S3** card to open the right pane, then configure the following settings under the **Data source properties - S3** tab:
 - For **Name**, enter **Source S3 bucket**.
 - For **S3 source type**, choose **Data Catalog table**.
 - For **Database**, use the dropdown menu and choose **weather_data**.
 - For **Table**, use the dropdown menu and choose **data**.

Task 4.2: Configure the data transformation

53. On the top-left side of the canvas, choose the blue **+** button to open the **+ Add nodes** pane.
54. Choose the **Transforms** tab, then choose **Change Schema**.

A **Change Schema** card is placed on the canvas and connected to the data source.

55. Choose the **Change Schema** card to open the right pane, and then configure the following settings under the **Change Schema (Apply mapping)** section:
- Change the name of the **date** attribute.
 - Under **Target key**, replace *date* with **report_date**.
 - Change the **date** attribute **Data type**:
 - From the **Data type** drop-down list, choose **date**.
 - Remove all attributes that have a **long** data type.
 - For each column with a **long** data type, under **Drop**, select the check box.

Note: These are the columns that should be removed from the dataset:

- temp_attributes
- dewp_attributes
- slp_attributes
- stp_attributes
- visib_attributes
- wdsp_attributes
- frshtt

56. Review attributes and their data types. Ensure the **date** data type was updated and renamed. Verify that all long columns have been excluded from the data transformation output.

Task 4.3: Define the job target and the output details for the job

57. On the top-left side of the canvas, choose the blue **+** button to open the **+ Add nodes** pane.
58. Choose the **Targets** tab, then choose **Amazon S3**.

An **Amazon S3** card is placed on the canvas and connected to the **Change Schema** card.

59. Choose the newly created **Amazon S3** card to open the right pane, then configure the following settings under the **Data target properties - S3** tab:

- For **Name**, enter **Target S3 bucket**.
- For **Format**, use the dropdown menu and choose **Parquet**.
- For **Compression type**, use the dropdown menu and choose **Uncompressed**.
- For **S3 Target Location**, choose **Browse S3**, then select the radio button next to the bucket that begins with **glue-bucket-**, then choose **Choose**.
- For **Data Catalog update options**, select the radio button next to **Create a table in the Data Catalog and on subsequent runs, update the schema and add new partitions**.
- For **Database**, use the dropdown menu and choose **weather_data**.
- For **Table name**, enter **data_parquet**.
- Choose **Add a partition key**, then for **Partition (0)**, use the dropdown menu and choose **report_date**.

Task 4.4: Complete the job configuration and save the job

60. Above the canvas, choose the **Job details** tab and configure the following settings:

- For **Name**, enter **DataPreparation**.
- For **IAM role**, use the dropdown menu and choose **gluelab**.
- Expand the **Advanced properties** section, and for **Script filename**, enter **data_prep.py**.

61. Above the canvas, choose the **Script** tab.

Take a moment to review the Python code that was automatically generated using the visual editor. This method for creating the job provides a helpful jump start for creating extract, transform, and load (ETL) code. You can update the Python script, if needed, before you save and run your ETL job. For this job, you do not need to make any changes, so you only save the job and continue to the next step.

62. In the upper-right corner, choose **Save**.

Note: You should see the following success message on top of the screen:

Successfully updated job

Successfully updated job DataPreparation. To run the job choose the Run Job button.

Task 4.5: Start the job

63. Choose **Run** to start your job.

64. Above the canvas, choose the **Runs** tab.

Notice that your job's **Run status** is *Running*.

Note: This job runs for 15-20 minutes.

You now know how to cleanse and transform data using an AWS Glue Studio job. Great work!

While the **DataPreparation** job is running, review the following helpful tips for managing AWS Glue crawlers and ETL jobs:

- [Why is the AWS Glue crawler running for a long time?](#)
- [Best practices to optimize cost and performance for AWS Glue streaming ETL jobs](#)

65. Verify that the job completed successfully before you continue to the next task.

Refresh: You may need to refresh the jobs under the **Runs** tab to check the status of your job. After successful run, the run status should move to **Succeeded**.

Task complete: You have successfully created and run an AWS Glue Studio job to clean up the data.

Task 5: Query the data_parquet table in Amazon Athena

In this task, you query the data using Amazon Athena.

Amazon Athena is an interactive query service that you can use to analyze data in Amazon S3 using standard SQL. Athena is serverless, so there is no infrastructure to manage, and you pay only for the queries that you run.

Learn more: Refer to *Amazon Athena Documentation* in the **Additional resources** section for more information.

66. At the top of the AWS Management Console, in the search bar, search for and choose **Athena**.

67. On the Amazon Athena **Get started** page, choose **Launch query editor**.

Note: If a **Workgroup primary settings** window opens, choose **Acknowledge**.

68. Choose the **Settings** tab.

69. Choose **Manage**.

70. On the **Manage settings** page, in the **Query result location and encryption** section, for **Location of query result - optional**, choose **Browse S3**.

71. On the **Choose S3 data set** pop-up window, select the radio button next to the bucket the begins with **query-result-**.

72. Choose **Choose**.

Note: Before you run a query in Amazon Athena, you must first specify the query result output location.

73. Choose **Save**.

74. Choose the **Editor** tab.

75. In the **Data** pane, configure the following settings:

- For **Data source**, use the dropdown menu and choose **AwsDataCatalog**.
- For **Database**, use the dropdown menu and choose **weather_data**.

Under **Tables and views**, you find two tables: **data**, and **data_parquet**. You query the **data_parquet** table.

Note: If you do not find a table named **data_parquet**, ensure that the AWS Glue job has completed. If the job has completed, refresh the browser page and the table should appear.

76. Next to the table named **data_parquet**, choose the ellipsis icon.

77. Under **Run Query**, choose **Preview Table**.

A new query tab opens, and the following query runs automatically:

```
SELECT * FROM "weather_data"."data_parquet" limit 10;
```

In the lower pane, notice a preview of the first 10 rows that the query retrieved.

78. To the right of the query tab, choose the **+** symbol, then enter the following SELECT statement:

```
SELECT report_date, latitude, longitude, temp, prcp
FROM "weather_data"."data_parquet"
WHERE report_date between CAST('2022-01-01' AS DATE) and
CAST('2022-06-30' AS DATE)
ORDER BY report_date
```

79. Choose **Run**.

This query returns the location, temperature, and precipitation data that was reported between January and June of 2022.

80. Replace the previous SELECT statement with the following SELECT statement:

```
SELECT report_date, avg(temp) as "Avg Temp", avg(prcp )
as "Avg Prcp"
FROM "weather_data"."data_parquet"
WHERE report_date between CAST('2022-01-01' AS DATE) and
CAST('2022-09-30' AS DATE)
Group by report_date
Order by 3 desc
```

81. Choose **Run**.

This query shows the average temperature, and average precipitation for each report_date between January 1, 2022 and September 30, 2022. The data returned is ordered so the wettest day during this time period appears first. Which report_date did your query return first?

As next steps, you could continue to explore the dataset, or you could hand access to your data science team so they can discover trends in the data. You could use this dataset to enrich other datasets that are already in use. You might also consider including partitions that you originally filtered out during your POC to provide more historic data to your data science team.

Task complete: You have successfully run queries to the data using Amazon Athena. In a very short time, you were able to access, prepare, and explore the NOAA GSOD dataset.

Conclusion

You have successfully done the following:

- Created an AWS Glue crawler.
- Created and ran a job in AWS Glue Studio.
- Explored permissions required to run AWS Glue crawlers and AWS Glue Studio jobs.
- Queried the AWS Glue Data Catalog using Amazon Athena.

End lab

Follow these steps to close the console and end your lab.

82. Return to the **AWS Management Console**.
83. At the upper-right corner of the page, choose **AWSLabsUser**, and then choose **Sign out**.
84. Choose **End Lab** and then confirm that you want to end your lab.

Additional resources

- [AWS Glue Documentation](#)
- [Amazon Athena Documentation](#)
- [Include and exclude patterns](#)

For more information about AWS Training and Certification, see <https://aws.amazon.com/training/>.

Your feedback is welcome and appreciated.

If you would like to share any feedback, suggestions, or corrections, please provide the details in our [AWS Training and Certification Contact Form](#).